

Infrastructure Cost Calculation

Original Requirement (from Project Documentation)

Calculation of Infrastructure Cost

This will be fully totally automated based on the Gross Land Area (GLA) entered during the masterplan initiation multiplied by the automatically selected rate of either low, mid, or high density.

There is in-built capability within the platform to enable users (with the appropriate access rights) to reconfigure the FAR density ranges and update the infrastructure cost models accordingly.

The total FAR that will identify the density is calculated based on the total FAR for each building asset.

How Infrastructure Cost Works

Infrastructure cost is **fully automated** - the platform calculates it automatically based on:

- 1. The **Gross Land Area (GLA)** entered when creating the masterplan
- 2. The **density category** determined by the total FAR from all building assets
- 3. The **infrastructure rate** configured for each density level

AUTOMATIC CALCULATION
Infrastructure Cost = Gross Land Area × Density Rate
The density rate is automatically selected based on the FAR calculated from all building assets in the masterplan.

Density Categories

The platform automatically determines which density category applies based on the total FAR:

FAR DENSITY RANGES
FAR < 0.465
— LOW DENSITY
Suburban-style development, lower infrastructure needs
FAR 0.465 to 1.5
— MEDIUM DENSITY
Mixed residential/commercial, moderate infrastructure
FAR > 1.5

└─ HIGH DENSITY
Urban high-rise development, highest infrastructure needs

Density Category	FAR Range	Description
Low	$\text{FAR} < 0.465$	Lower infrastructure requirements
Medium	$0.465 \leq \text{FAR} < 1.5$	Moderate infrastructure requirements
High	$\text{FAR} \geq 1.5$	Higher infrastructure requirements

Calculation Flow Diagram

FROM MASTERPLAN

Gross Land Area (GLA) = Entered during masterplan creation



FROM BUILDING ASSETS

Total GFA = Sum of all building assets GFA
Total Plot Area = Sum of all building assets plot areas



STEP 1: CALCULATE FAR

Total FAR = Total GFA ÷ Total Plot Area
This FAR is calculated from ALL building assets combined



STEP 2: DETERMINE DENSITY

FAR < 0.465 → LOW DENSITY

$0.465 \leq \text{FAR} < 1.5 \rightarrow \text{MEDIUM DENSITY}$
 $\text{FAR} \geq 1.5 \rightarrow \text{HIGH DENSITY}$



STEP 3: GET RATE

Infrastructure Rate (SAR per m²) = Retrieved from Cost Model
based on density category



STEP 4: CALCULATE COST

Net Infrastructure Cost = Gross Land Area × Rate per m²

General Requirements Amount = Net Cost × (Gen Req % ÷ 100)

Total Infrastructure Cost = Net Cost × (1 + Gen Req % ÷ 100)



STEP 5: CALCULATE SPLIT

Infrastructure is split into Primary and Secondary:

Primary Infrastructure (30%) = Roads, main utilities, trunk lines

Secondary Infrastructure (70%) = Distribution networks, connections



OUTPUT

- Calculated FAR
- Density Category (Low/Medium/High)
- Rate per m² GLA
- Net Infrastructure Cost (SAR)
- Primary Cost (30%)
- Secondary Cost (70%)
- General Requirements Amount (SAR)
- Total Infrastructure Cost (SAR)

- Balance External Area (GLA – Total Plot Area)

Example Calculation

Masterplan Input:

- Gross Land Area: 500,000 m²
- General Requirements: 10%

Building Assets Totals:

- Total GFA from all buildings: 750,000 m²
- Total Plot Area from all buildings: 400,000 m²

Step 1 - Calculate FAR:

- $\text{FAR} = 750,000 \div 400,000 = \mathbf{1.875}$

Step 2 - Determine Density:

- $\text{FAR } 1.875 \geq 1.5 \rightarrow \mathbf{\text{HIGH DENSITY}}$

Step 3 - Get Rate:

- High Density Rate (from Cost Model): **350 SAR per m²**

Step 4 - Calculate Cost:

- Net Infrastructure Cost = $500,000 \times 350 = \mathbf{175,000,000 \text{ SAR}}$
- General Requirements = $175,000,000 \times 0.10 = \mathbf{17,500,000 \text{ SAR}}$
- Total Infrastructure Cost = $175,000,000 + 17,500,000 = \mathbf{192,500,000 \text{ SAR}}$

Step 5 - Calculate Split:

- Primary Infrastructure (30%) = $175,000,000 \times 0.30 = \mathbf{52,500,000 \text{ SAR}}$
- Secondary Infrastructure (70%) = $175,000,000 \times 0.70 = \mathbf{122,500,000 \text{ SAR}}$

Balance External Area:

- Balance = $500,000 - 400,000 = \mathbf{100,000 \text{ m}^2}$ (available for roads, utilities, etc.)

What Infrastructure Includes

Standard infrastructure costs cover:

Category	Items Included
Roads	Main roads, internal roads, paving
Drainage	Storm water, sewerage networks
Water Supply	Water distribution networks, connections
Electrical	Power distribution, street lighting
Telecommunications	Network infrastructure

Landscaping	Basic landscaping allowances
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Configuration Options

Users with appropriate access can configure:

1. **FAR Density Thresholds** - Adjust the FAR ranges that define Low/Medium/High density
 2. **Infrastructure Rates** - Update the SAR per m² rate for each density level
 3. **Infrastructure Split** - Modify the 30/70 split between primary and secondary infrastructure
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Balance External Area

The platform automatically calculates the Balance External Area:

Balance External Area = Gross Land Area - Total Building Plot Areas

This represents the land available for:

- Roads and circulation
- Utilities and infrastructure
- Public spaces not allocated to public realm
- Buffer zones

If this value becomes negative, it indicates that building plot areas exceed the available land.